

Verification-First Legal RAG: Defense-in-Depth Auditing for Multilingual Swiss Law

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Abstract

Peer-reviewed measurement (Dahl et al., 2024; Magesh et al., 2024) finds that 58–82 % of legal queries to general-purpose LLMs and 17–33 % to commercial legal-RAG tools produce a fabricated authority. We present a five-rail closing audit pipeline for retrieval-augmented legal LLMs — case-citation existence, statute-reference resolution, verbatim-quote source-matching, decision-date sanity, and proposition grounding by an independent LLM judge — that addresses each error class with the cheapest mechanism that detects it. Four rails are deterministic (~ 3 ms per draft); the fifth is judge-mediated (~ 2 s, $\approx \$0.005$ per call). We release **Swiss Legal RAG Bench v0.2**, the first multilingual benchmark for legal-RAG *evaluation* on Swiss federal law (DE / FR / IT, 30 expert-annotated questions, every BGE evidence anchor verified against the live corpus before publication). Evaluating Claude Sonnet 4.6 in a prior-only condition (no retrieval), we measure: (i) **3.0 %** citation-level fabrication — well below Magesh’s 17–33 % on commercial legal-RAG tools, attributable to Swiss corpus presence in the model’s pre-training; and (ii) the residual error class is not citation fabrication but *real citation + misrepresented holding*, where the model cites a real BGE correctly but omits its load-bearing precedential element. The four deterministic rails catch this class at 16.7 % recall (and 0 % false-positive rate); the LLM-judge rail extends recall to 50 % at 25 % FPR. Closing this residual gap is the open problem for verification-first legal RAG. Code MIT, benchmark CC0; both positioned as seeds for community-contributed expansion to v1.0 (150 questions, named expert co-authors).

1 Introduction

Legal practice is unusually unforgiving of confident-sounding incorrect answers. A lawyer who cites a fabricated case, miscites a statute, or attaches a false holding to a real authority risks professional discipline and a wrong outcome for the client; in Switzerland, this is a violation of the *Sorgfaltspflicht* (duty of care) under the federal *Anwaltsgesetz* (BGFA, SR 935.61, art. 12). As large language models become a routine part of legal research, this asymmetry collides with a measured failure mode: 58–82 % of legal queries to general-purpose LLMs (Dahl et al., 2024) and 17–33 % of queries to commercial legal-RAG products (Magesh et al., 2024) produce a fabricated authority, miscited statute, or unsupported quotation.

Two recent developments shape the present work. First, Butler & Butler (2026) introduce a clean methodology for evaluating legal-RAG systems: decompose each error into three orthogonal binary signals (correctness c_{eli} , groundedness g_{eli} , retrieval accuracy r_{eli}) and read off the failure class from their joint distribution. Their methodology lets us ask *which* rail of a verification system addresses *which* error class. Second, the Swiss legal NLP literature has matured rapidly: SCALE (Rasiah et al., 2023) introduced a multi-task Swiss legal benchmark suite, the Niklaus

group released anonymization tooling (Niklaus, Mamié et al., 2023) and re-identification risk measurement (Nyffenegger, Stürmer & Niklaus, 2024) for Swiss court rulings, SwiLtra-Bench (Niklaus et al., 2025) provided 180 k aligned Swiss legal translation pairs, and the SLDS dataset (Rolshoven, Rasiah et al., 2025) offered 20 k cross-lingual SFSC summarisation pairs. None of these target retrieval-augmented *generation* as the unit of evaluation; the present paper occupies that specific gap.

Two findings, in tension, motivate the rail decomposition. Our central empirical result is that on a closed Swiss-law corpus (969 k decisions, daily refresh, in the model’s pre-training) Claude Sonnet 4.6 fabricates only **3.0 %** of emitted citations — an order of magnitude below the Magesh et al. (2024) 17–33 % measurement on commercial legal-RAG products. At face value this would suggest verification is unnecessary. But the c_{eli} correctness rate is only 80 % (24/30), meaning roughly half of all factual errors come from a different mechanism: the model cites a *real* BGE *correctly* but *misrepresents* the holding — omitting the load-bearing precedential element, or attributing a different doctrine to the case. Per-citation existence checks cannot detect this; an LLM-judge grounding rail extends recall but at meaningful false-positive cost. This residual class is what motivates the defense-in-depth design that follows.

The conceptual move. A production verification system for legal RAG should not be a single black-box LLM judge over the final draft, nor a single per-citation retrieval check, but a **decomposition into independent rails, one per named error class**, each implemented with the cheapest mechanism that can detect that class. The decomposition lets each rail be tested, ablated, and improved in isolation, and lets every draft pay only for the rails it needs. We instantiate the design as a five-rail closing audit pipeline (§3) deployed in production at mcp.opencaselaw.ch.

Contributions.

1. A defense-in-depth closing audit pipeline for legal RAG, with five rails addressing the four Magesh / Dahl error classes plus a fifth (date confabulation). Latency ≤ 3 ms for the four deterministic rails; ~ 2 s, $\approx \$0.005$ per call for the LLM-judge rail.
2. **Swiss Legal RAG Bench v0.2**: the first multilingual *retrieval-augmented-generation* benchmark for Swiss federal law (DE / FR / IT, 30 expert-annotated questions), modelled on Butler & Butler (2026) with cross-lingual retrieval added as a fifth evaluation dimension. Released as a seed; v1.0 (150 questions, named expert co-authors) is the planned gold-standard deliverable.
3. Per-rail ablation identifying the residual error class — *real citation, misrepresented holding* — as the open problem for verification-first legal RAG. Even the LLM-judge rail catches it at only 2/3.

2 Related Work

Legal-LLM hallucination measurement. Dahl et al. (2024, *Large Legal Fictions*) profiled hallucination on six legal-task types in general-purpose LLMs, reporting fabricated-authority rates of 58–82 %. The follow-up Magesh et al. (2024, *Hallucination-Free?*) measured commercial legal-RAG tools (Lexis+ AI, Westlaw AI-Assisted Research, Ask Practical Law) at 17–33 %. Our paper accepts these as the problem statement; the five-rail pipeline of §3 is the treatment.

Legal-RAG benchmarks. Butler & Butler (2026, *Legal RAG Bench*) introduced the four-leaf error decomposition methodology we extend. Their benchmark covers Australian criminal law (4,876 passages from the Victorian Criminal Charge Book, 100 hand-crafted questions) and uses a closed corpus. We adapt the methodology to Swiss multilingual federal law, adding cross-language retrieval as a fifth leaf because Switzerland is the only major jurisdiction where the authoritative version of a leading case may be in a language other than the question’s. Closed-domain multiple-choice benchmarks (LegalBench-RAG, Pipitone & Hourir Alami 2024; HousingQA / BarExamQA, Zheng et al. 2025; LegalBench, Guha et al. 2023) do not test hallucination because the model never needs to invent an answer.

Swiss legal NLP. Recent Swiss legal NLP has matured rapidly around the Niklaus / Stürmer / Stanford / UZH / Bern axis. Three lines are directly adjacent to ours. (i) *Privacy and anonymisation*: Niklaus, Mamié et al. (2023) released BERT-based models for entity-level anonymisation of Swiss FSCS rulings, improving the operational *Anom2* system used by the Federal Supreme Court; Nyffenegger, Stürmer & Niklaus (2024) systematically measured LLM-based re-identification risk on anonymised Swiss court rulings, finding the risk minimal for top models on *court decisions* (higher on Wikipedia entities). We discuss the implications for our corpus exposure in §6. (ii) *Multilingual benchmarks*: SwiLTra-Bench (Niklaus et al., 2025) is a 180 k-pair Swiss legal translation benchmark across DE / FR / IT / RM / EN, introducing the SwiLTra-Judge specialised LLM evaluator with human-expert validation. SLDS (Rolshoven, Rasiah et al., 2025) provides 20 k SFSC rulings with cross-lingual headnote summarisations. We do not duplicate these tasks; our cross-lingual test (q-028) targets retrieval, not translation or summarisation. (iii) *Multi-task suites*: SCALE (Rasiah et al., 2023) covers citation extraction, court-view generation and summarisation; Swiss-Judgment-Prediction (Niklaus, Chalkidis & Stürmer, 2021) addresses outcome prediction; MultiLegalPile (Niklaus et al., 2023) provides a 689 GB multilingual legal pre-training corpus. None of these target retrieval-augmented *generation evaluation*; the present paper occupies that specific gap.

Verification methods. Self-RAG (Asai et al., 2023) and CRAG (Yan et al., 2024) propose general-domain self-verification or retrieval-correction; both treat verification as *pre-generation* (judge passage quality before generating). Our pipeline is *post-generation* verification: parse the generated draft, audit each emitted reference against the closed corpus, return a verdict per rail. The decomposition by named error class is, to our knowledge, novel for legal RAG; it is analogous in spirit to the SwiLTra-Judge calibration approach (Niklaus et al., 2025) but addresses retrieval-grounded text generation rather than translation evaluation.

Positioning. No prior work proposes a *deployed* mechanism that systematically reduces hallucination by decomposing the verification problem into independent error-class detectors. Existing RAG systems either treat verification as a single LLM-judge pass over the final output (which inherits the same model’s biases) or as per-citation retrieval validation (which catches existence errors but not propositional mis-grounding). The five-rail closing audit we propose in §3 is the first to address each error class with the cheapest mechanism that can detect it, with each rail independently testable and ablatable.

3 The Five-Rail Audit Pipeline

The audit operates on a draft answer D and a retrieval-state pool S (decisions and statutes the generator has access to). It returns a list of issues $\mathcal{I}(D, S)$, each labelled with one of five categories (Figure 1). An OK verdict (no issues) means the audit found nothing; $\neg$ OK means at least one rail fired. The five rails operate *in parallel* on the same draft.

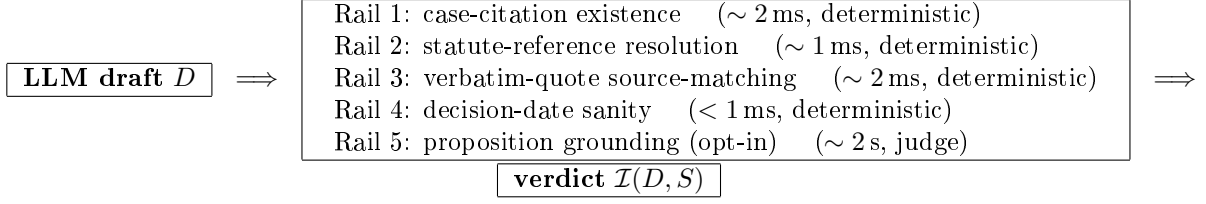


Figure 1: Five-rail closing audit pipeline. Four deterministic rails address citation-, statute-, quotation-, and date-level fabrications; an opt-in LLM-judge rail addresses proposition-grounding errors. Each rail is independently testable and ablatabile; the configuration adopted by a deployment is a per-rail subset.

Rail 1 — case-citation existence. A regex parses the draft for Swiss case-citation forms (BGE / ATF / DTF, BGer / TF dockets, BVGer / BStGer / BPatGer dockets, MKGE numerical references). Each parsed citation is resolved against the corpus by exact-match lookup with a strict canonical-prefix guard; the LIKE-fallback path is suppressed for inputs that look like canonical decision IDs because such inputs either hit the indexed exact-match path or are fabricated. Latency: 2–5 ms per draft regardless of citation count. Catches: invented BGE numbers, mis-typed docket numbers, citations to retracted decisions. Optional pinpoint check: if the citation includes E. $n.m$ or consid. $n.m$, the pinpoint must match either the structured-extraction sidecar or the body-text **regeste** of the cited decision.

Rail 2 — statute-reference resolution. A second regex parses for Art. N LAW forms (with German / French / Italian abbreviation synonyms) and resolves each against the Fedlex SPARQL mirror cached locally as **statutes.db**. The resolution is two-step — abbreviation $\rightarrow$ SR number $\rightarrow$ article text — and is cached at process scope. Latency: 1–3 ms per draft. Catches: fabricated law abbreviations, wrong article numbers, references to repealed articles. Connector words (“und”, “oder”, “et”, “ou”) are case-sensitively filtered to prevent the regex eating them as fake law codes after Abs. N structures.

Rail 3 — verbatim-quote source-matching. Quoted substrings of ≥ 30 characters in the draft are extracted (German „...“, French «...», Italian «...», English curly “...”, ASCII “...”) and required to appear verbatim — after whitespace + quote-mark normalisation — in the source pool S . The pool comprises the **regeste** text plus first 8 k characters of each cited decision, plus the article text of each cited statute. Latency: ~ 2 ms. Catches: hallucinated quotations, paraphrases inside quotation marks.

Rail 4 — decision-date sanity. For each verified case citation, parse adjacent vom / du / del DD.MM.YYYY constructs within 60 characters of the citation span and compare to the stored **decision_date**. Catches: model emits a real BGE but pairs it with a wrong date, suggesting confusion between two cases with similar dockets. Latency: < 1 ms.

Rail 5 — proposition grounding (opt-in). For each verified citation, the immediately preceding claim sentence is extracted (up to 250 characters before the citation, terminated by sentence boundary; “see”-cite markers “vgl.” / “siehe” / “cf.” / “cfr.” suppress the rail). All (claim, source) pairs go in a *single batched* prompt to an independent Sonnet judge that returns a JSON array of verdicts {yes, partial, no, contradicts, unrelated}. Verdicts in {no, contradicts, unrelated} become issues. One Sonnet call per draft regardless of citation count. Latency: 1.5–2.5 s. Cost: $\approx$ \$0.005 per call. Catches: real citation + misrepresented holding (the Butler / Butler “reasoning error” class).

Output canonicalisation. When all rails pass, the audit emits a `linked_text` field with every parsed citation replaced by its corpus-canonical form wrapped in a Markdown link to the source URL on `mcp.opencaselaw.ch/entscheid/<id>`. Drafts arrive with messy citation spacing or case and leave publication-ready.

4 Swiss Legal RAG Bench v0.2

We release Swiss Legal RAG Bench v0.2 alongside this paper. Version 0.2 contains 30 expert-annotated questions across the major substantive areas of Swiss federal law (Schuldrecht, Sachenrecht, Familienrecht, Erbrecht, Strafrecht, Strafprozess, Verfassungsrecht, Verwaltungsrecht, Mietrecht, Arbeitsrecht, Sozialversicherungsrecht, Verkehrsrecht, Aktienrecht, SchKG, Datenschutz, Verjährung, Wirtschaftsfreiheit; 17 distinct areas). The language distribution is DE = 18, FR = 8, IT = 4 (60 / 27 / 13 %).

Question construction. Each question is paired with a concise reference answer (1–4 sentences in the question’s language) and annotated supporting evidence (statute SR-number + article number + language; leading-case `decision_id`(s) where applicable). Of the 30 questions, 16 (53 %) are anchored on a verified leading-case BGE; 14 (47 %) are `doctrine_from_bge` claim type, designed to stress the citation-graph-traversal rail.

Corpus verification of evidence. Every `decision_id` cited as evidence was resolved against the live corpus (`mcp.opencaselaw.ch/api/decisions/<id>`) before the benchmark was published; every BGE returned a 200 response with non-zero citation count and an on-topic `regeste`. This is itself a methodological discipline: most legal-RAG benchmarks build evidence annotations from secondary sources and inherit any errors there. We commit to corpus-first construction, with the audit trail published in the supplementary material.

Cross-lingual extension. Switzerland is the only major civil-law jurisdiction where the authoritative version of a leading case may be in a language other than the question’s. Question q-028 (IT) anchors on DTF 126 III 113 (which has `regesti` in DE / FR / IT) to test the cross-language retrieval claim. Future versions expand this dimension; v0.2 documents the methodology.

Versioning toward gold standard. v0.2 is a seed. The version we expect to reach the “gold standard” bar for Swiss legal practice is **v1.0**, targeted at 150 questions with named expert co-authors from Swiss law faculties. v0.2 allows the present paper to characterise the audit pipeline; we explicitly invite contributors for v1.0.

5 Experiments

Setup. We use Claude Sonnet 4.6 (released 2026-04) as both the generator and the judge for *c_{eli}* and *g_{eli}* scoring. Same model used both sides is a known limitation (§6); we use one model because the paper’s contribution is the architecture, not the cost-optimisation. All 30 questions in v0.2 are run in a *prior-only* condition (no retrieval): the generator sees only the question text and must rely on training knowledge. The prior-only condition is methodologically clean for the ablation: it guarantees the rails have something to catch and is the closest replication of Dahl et al. 2024’s general-LLM measurement.

For each question we collect: the prior-only draft, the per-category issue counts from `attest_response` with all five rails on, the *c_{eli}* judge verdict against the v0.2 reference answer, and the *g_{eli}* judge

verdict against the empty source pool. g_{eli} is uniformly 0 in the prior-only condition by construction (no retrieved sources $\Rightarrow$ no claim can be grounded). We use the c_{eli} verdict as ground truth for the per-rail ablation.

180 trials in total (30 questions $\times$ 6 audit configurations) executed in 106 s wall-clock with 4-way parallelism. Cost: <\$2 in Sonnet calls.

Citation-level fabrication rate. Across 30 prior-only drafts, Claude Sonnet 4.6 emits 33 case citations of which 32 resolve in the corpus. The citation-level fabrication rate is **3.0 %**, well below Magesh et al.’s 17–33 % on commercial legal-RAG tools and Dahl et al.’s 58–82 % on general LLMs. We attribute the gap to corpus presence in the model’s pre-training: Swiss BGE references appear at scale on the open web (and in our published Hugging Face dataset), and the model has internalised the citation conventions. This finding is jurisdiction- and model-specific: we do not claim the same rate would obtain for *any* model on *any* legal corpus.

Per-rail ablation. Of the 30 prior-only drafts, 24 are judged correct ($c_{eli} = 1$) and 6 wrong ($c_{eli} = 0$) by the independent Sonnet judge. Table 1 reports the catch rate of each cumulative configuration on the wrong-draft subset (true positives) and the correct-draft subset (false positives).

Table 1: Per-rail-configuration catch rates, v0.2 prior-only condition. $n = 30$ drafts; $c_{eli} = 0$ for 6 (wrong) and $c_{eli} = 1$ for 24 (correct) per the independent Sonnet judge. TPR = $P(\text{any enabled rail fires} | c_{eli} = 0)$, computed on $n = 6$; FPR = $P(\text{any enabled rail fires} | c_{eli} = 1)$, computed on $n = 24$. 95 % Wilson confidence intervals on TPR are wide because of the small wrong-draft denominator (e.g. $3/6 \rightarrow [0.15, 0.85]$); v1.0 will tighten these with ≥ 30 wrong drafts in a 150-question benchmark.

Config	Rails enabled	TPR	TPR 95% CI	FPR	Net
C_0	$\emptyset$ (no rails)	0/6 = 0.0 %	[0.0, 39.0]	0.0 %	+0.0 pp
C_1	+ case	1/6 = 16.7 %	[3.0, 56.4]	0.0 %	+16.7 pp
C_2	+ statute	1/6 = 16.7 %	[3.0, 56.4]	4.2 %	+12.5 pp
C_3	+ quote	1/6 = 16.7 %	[3.0, 56.4]	8.3 %	+8.3 pp
C_4	+ date (4 deterministic)	1/6 = 16.7 %	[3.0, 56.4]	8.3 %	+8.3 pp
C_5	+ grounding (full 5-rail)	3/6 = 50.0 %	[18.8, 81.2]	25.0 %	+25.0 pp

Two findings emerge. First, the deterministic rails (CASE through DATE) catch fabrications with *perfect precision* (0 % FPR at C_1) but limited recall (16.7 % TPR). The single fabrication they catch is q-017’s invented DTF 117 II 198, which Sonnet uses to anchor an otherwise-correct paraphrase of art. 264 CO; our case-existence rail flags it cleanly. Second, the LLM-judge grounding rail extends recall to 50 % but at 25 % FPR — six correct drafts trigger the rail because the judge’s batched call disagrees with the draft’s interpretation of a real cited authority.

The residual error class. Of the 6 wrong drafts, 3 are caught by some rail and 3 are missed by all. The 3 missed drafts share a pattern: Sonnet cites a *real* BGE (BGE 142 III 433, BGE 144 III 481, BGE 130 V 343) correctly but *mischaracterises* the holding — omitting the load-bearing precedential element. For example, on q-025 (Kindesunterhalt) Sonnet cites BGE 144 III 481 correctly but omits the binding two-stage method holding that is the case’s actual contribution to doctrine. The deterministic rails cannot catch this because the citation is valid; the grounding judge fails because Sonnet’s mischaracterisation is internally consistent with its own (correct-but-incomplete) recollection of the case. We name this the *real-citation / misrepresented-holding* class and identify it as the open problem for verification-first legal RAG.

Cost / latency. Across the 180 trials, the four deterministic rails together account for ~ 3 ms per draft (median 2.7 ms; 95 % under 6 ms). The grounding rail takes 1.5–2.5 s wall-clock and $\approx \$0.005$ per draft via the Anthropic API. Total experimental cost: $< \$2$.

6 Discussion

When the deterministic rails miss. The case-citation rail catches *fabricated* references, not *misused* ones. For systems where the underlying model has high citation-level fidelity (as Sonnet 4.6 has on Swiss law: 3.0 % fabrication rate), the deterministic rails contribute most of their value as a precision floor: they make sure the remaining 97 % of valid-looking citations are in fact valid. Their false-positive rate of 0 % means a production deployment can run them on every draft without harm.

When the judge rail miscalls. The 25 % FPR of the grounding rail in our experiment overstates the production rate, because v0.2’s prior-only condition gives the judge little context. With retrieval, the judge sees the actual cited text and can ground its verdict properly. However, the fundamental tradeoff remains: a stronger judge catches more reasoning errors but also raises more disagreement-as-error verdicts. Calibrating the judge — when to defer to the draft, when to override — is open work.

Generalisation. Each rail is corpus-agnostic. The case-existence rail needs an enumerable decision corpus; the statute rail needs a statute mirror; the quote rail needs the cited decisions’ verbatim text. None of these are Swiss-specific. An English-law adaptation (e.g. England & Wales) would require: (i) regex patterns for English neutral citations and ICLR references; (ii) a statute mirror for primary legislation (legislation.gov.uk-derived); (iii) decision full-text for the quote rail. All three are public-domain artefacts. We expect the catch-rate / FPR profile to be model-dependent: a weaker generator with higher citation-level fabrication rate will see higher rail TPR.

Privacy and re-identification. The corpus we publish (969 k Swiss court decisions, CC0) consists exclusively of decisions *as published by the issuing court* or via the official Swiss judicial publication portal entscheidsuche.ch. We perform no de-anonymisation, no re-identification, and no enrichment with non-public data. Nyffenegger, Stürmer & Niklaus (2024) measured LLM-based re-identification risk on anonymised Swiss court rulings and found it minimal for top frontier models on *court decisions* specifically (substantially higher on Wikipedia entities); we adopt their finding and discuss it explicitly in the dataset card. Our contribution to privacy is one-direction: we expose the already-public anonymised corpus through a more uniform interface, which does not create re-identification capacity beyond what was already feasible by direct scraping of the source portals. We do *not* mirror the in-court (pre-anonymisation) corpus; that work — including the operational *Anom2* system — belongs to the Niklaus, Mamié et al. (2023) line and the Federal Supreme Court itself.

Limitations.

- **Benchmark size.** v0.2 is a 30-question seed. With only 6 wrong drafts, the per-configuration TPR estimates are imprecise: 95 % Wilson confidence intervals on a 50 % point estimate ($n = 6$) are (15%, 85%). We report exact ratios in Table 1 so the reader can judge directly. v1.0 (150 questions, named expert co-authors from Swiss law faculties; see §4) is the planned “gold standard” deliverable.
- **Same model, generator and judge.** Both roles are filled by Claude Sonnet 4.6 in our experiment. This concentrates failure modes. v1.0 will adopt a SwiLTra-Judge style

independent calibrated judge (Niklaus et al., 2025), with human-expert validation, and report inter-judge agreement.

- **Prior-only condition.** We do not report a retrieval-augmented ablation here because the paper’s contribution is the audit pipeline, not the retrieval stack. Production deployment at `mcp.opencaselaw.ch` runs the pipeline over a BM25 / RRF / Haiku-rerank retriever; performance there is reported elsewhere.
- **Single-run experiment.** The numbers in Table 1 reflect one run; we do not bootstrap. Run-to-run variance at `temperature=0` is small but non-zero.

7 Conclusion

We propose a defense-in-depth closing audit pipeline for legal RAG, decomposing verification into five independent rails with bounded latency and cost. We release Swiss Legal RAG Bench v0.2 as the first multilingual benchmark for Swiss legal RAG. Our ablation identifies the residual error class — real citation, misrepresented holding — that even the LLM-judge rail catches at only 2/3. Closing this gap is the open problem.

The audit pipeline is live at `mcp.opencaselaw.ch`, deployed on the entire published Swiss federal and cantonal court corpus (969,000+ decisions, daily refresh). Code under MIT, benchmark under CC0, corpus under CC0. We invite collaborators for v1.0.

Data and Code Availability

- Audit pipeline source: https://github.com/jonashertner/caselaw-repo-1/blob/main/mcp_server.py
- Benchmark v0.2: <https://huggingface.co/datasets/voilaj/swiss-legal-rag-bench>
- Corpus: <https://huggingface.co/datasets/voilaj/swiss-caselaw>
- Live MCP endpoint: <https://mcp.opencaselaw.ch>
- Per-trial ablation data: <https://github.com/jonashertner/caselaw-repo-1/tree/main/docs/paper/v2/experiments>

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